
COMP 2121
DISCRETE
MATHEMATICS

Lecture 10



Sample Space:

1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

$$\Pr(\text{sum} \leq 9) = \frac{30}{36}$$

$$\Pr(\text{sum} \leq 9) = 1 - \Pr(\text{sum} > 9) = 1 - \frac{6}{36} = \frac{30}{36}$$

Relationship (Counting, Venn Diagrams)

Quiz 2 in Lab 5 (based on labs 3, 4)

Counting and Venn's diagrams

Recall that for a finite set A , $|A|$ denotes the number of elements of A and is referred to as the *cardinality*, or *size*, of A . For example, set $A = \{1, 2, 5, 23\}$ has a cardinality 4, i.e. $|A| = 4$.

Example 1. In a class of 50 college freshmen, 30 are studying C++, 25 are studying Java, and 10 are studying both languages. Draw Venn diagram and answer how many freshmen are studying C++ or Java. How many students are not studying any programming language?

C - students studying C++

J - " " " "

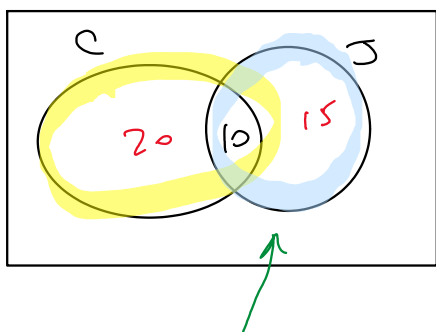
U - all students

$$|C| = 30$$

$$|J| = 25$$

$$|U| = 50$$

$$|C \cap J| = 10$$



$$\begin{aligned} |C \cup J| &= |C| + |J| - |C \cap J| \\ &= 30 + 25 - 15 \\ &= 45 \end{aligned}$$

$$|U| - |C \cup J| = 50 - 45 = 5$$

are not studying
any language.

Cardinality of the Union

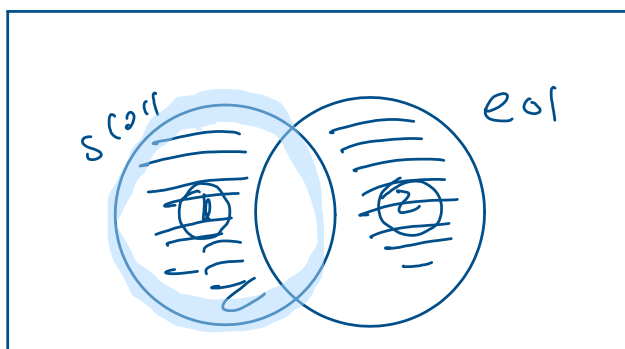
Theorem: If A and B are finite sets, then $|A \cup B| = |A| + |B| - |A \cap B|$. Consequently, finite sets A and B are mutually disjoint if and only if $|A \cup B| = |A| + |B|$.

Example 2. How many 8-bit strings start with 1011 or end with 01?

$$\begin{array}{l} \text{ST } 1011 \text{ or } \text{ET } 01 \\ 1011 \text{ --- } 2^4 \\ \text{or --- } 2^6 \therefore 2^4 + 2^6 \end{array}$$

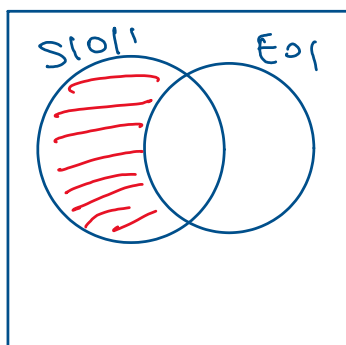
$$\begin{array}{l} \text{ST } 1011 \text{ or } \text{ET } 01 \\ 1011 \text{ --- } 2^4 \\ \text{--- --- } 2^6 \\ 1011 \text{ --- } 01 \quad 2^2 \end{array} \quad \underline{\underline{2^4 + 2^6 - 2^2}}$$

Example 3. How many 8-bit strings start with 1011 or end with 01, but not both?



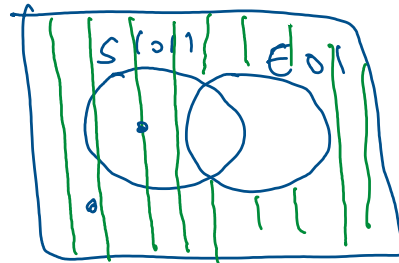
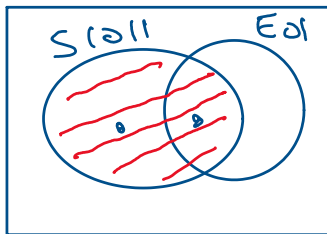
$$\textcircled{1} + \textcircled{2} \\ (2^4 - 2^2) + (2^6 - 2^2)$$

Example 4. How many 8-bit strings start with 1011 but does not end with 01?

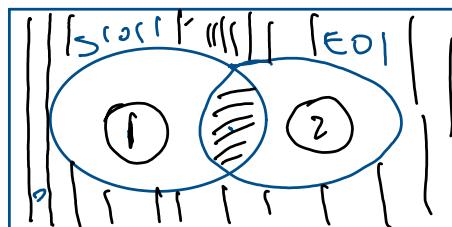


$$\begin{array}{l} 1011 \text{ --- } 2^4 \\ 1011 \text{ --- } 01 \quad 2^2 \\ 2^4 - 2^2 \end{array}$$

How many 8-bit strings start with 1011 or
does not end 01, but not both?



=



$$|A| - \textcircled{1} - \textcircled{2} = 2^8 - (2^4 - 2^2) - (2^6 - 2^2)$$

Theorem: *If A, B, C are finite sets, then*
$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

Example 5. A student visits an arcade each day after school and plays one game of either Pac-Man, Tetris, or Donkey Kong. In how many ways can he play one game each day so that he plays each of the three types at least once during a given school week?

Probability

Definition: *A sample space S is the set of all possible outcomes of a random process or experiment.*

For example, when rolling a single die the sample space is $S = \{1, 2, 3, 4, 5, 6\}$. Here we feel that each of six outcomes has the *same*, or *equal*, *likelihood* of occurrence.

The following definition is given under the assumption of equal likelihood:

Definition: *Let S be a sample space. Each subset A of S (including the empty set) is called an event. Each element of S determines an outcome, so if $|S| = n$ and $a \in S$, $A \subseteq S$, then:*

$$Pr(\{a\}) = \text{the probability that } \{a\} \text{ occurs} = \frac{1}{n}$$

$$Pr(A) = \text{the probability that } A \text{ occurs} = \frac{|A|}{|S|} = \frac{|A|}{n}$$

Observation: $Pr(S)=1$ and $Pr(\bar{A}) = 1 - Pr(A)$

Example 6. Dillon rolls a fair die, what is the probability he gets (a) an even number, (b) a number different from number 3?

$$\Pr(\text{even}) = \frac{3}{6}$$

$$\Pr(3) = \frac{1}{6} \quad \Pr(\bar{3}) = 1 - \Pr(3) \\ = 1 - \frac{1}{6} = \frac{5}{6}$$

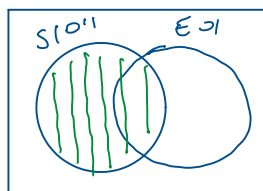
Example 7. What is the probability that a randomly selected 12-bit string starts with 1011 (or) ends with 011?

$$\Pr = \frac{2^8 + 2^4 - 2^5}{2^{12}}$$

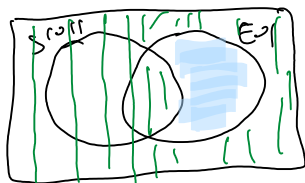
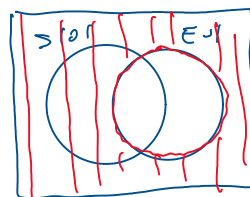
1011 ----
 ---- 011
 1011 ---- 011

$$\Pr = \frac{|A \cup B|}{|U|}$$

Find the probability that a randomly selected 12 bit string starts with 1011 or does not end with 011.

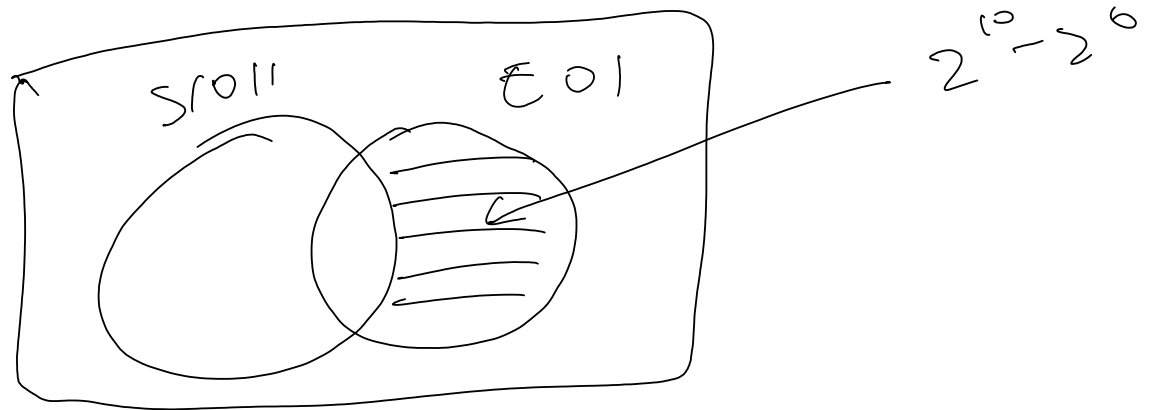


U



$$\Pr = \frac{2^{12} - (2^{10} - 2^6)}{2^{12}}$$

$$S1011 \text{ or } E01 = \neg \left[\overline{S1011 \text{ AND } E01} \right]$$



$$Pr \left[\overline{S1011 \text{ and } E01} \right] = \frac{2^{10} - 2^6}{2^{12}}$$

$$\text{answ: } 1 - \frac{2^{10} - 2^6}{2^{12}}$$

Independent events with different probabilities

So far we were dealing with the assumption of equal likelihood. However, that is not a general case. In general, events are not necessarily equally likely. We will restrict our discussion to independent events.

If two events are independent, then probability of $A \cap B$ is:

$$\Pr(A \cap B) = \Pr(A)\Pr(B).$$

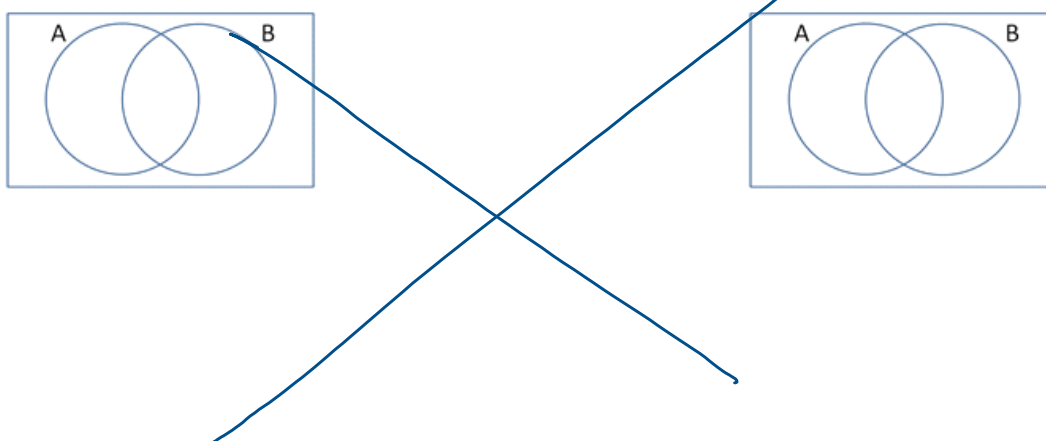
This is very intuitive, and it has analogy with the product rule. By the definition of intersection: $x \in A \cap B \Leftrightarrow x \in A \wedge x \in B$. The “key word” here is AND. We have learned that “AND” and multiplication are very much related.

Also, the probability of union is very much related to the counting argument presented for cardinality of the union:

$$\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B).$$

Example 8. Printer is working 80% of time and a photocopier is working 70% of the time.

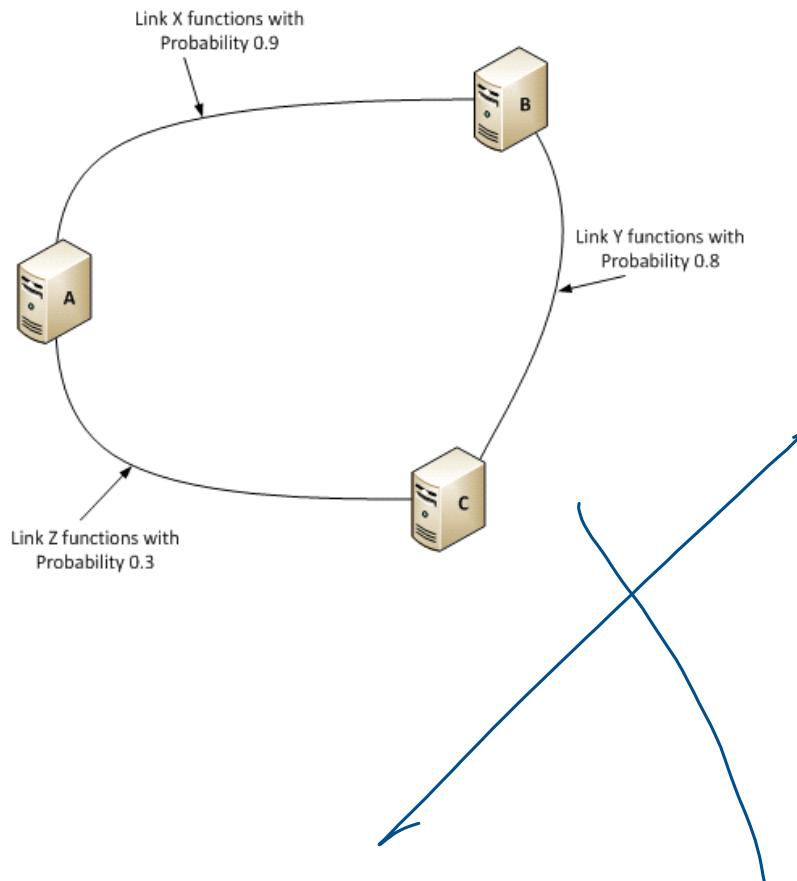
What percent of the time is at least one machine working?



What percent of the time is at least one machine down?

Example 9

Example 9. In the communication network shown below, each link may be up or down. Assuming, that the nodes connecting the links are always functioning and that failures of the links are not related, what is the probability that functioning set of links is connecting node A to node C?



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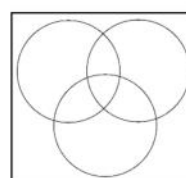
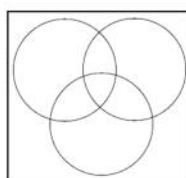
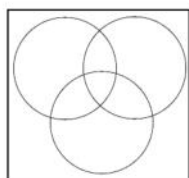
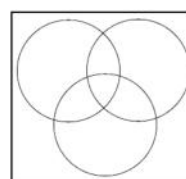
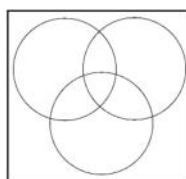
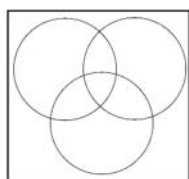
LAB 5

1. Let $X = \{1, 2, 3\}$. Which of the following statements are true?

- a) $\{3\} \in X$ b) $3 \in X$ c) $\{3\} \subseteq X$ d) $\{1, 3\} \subseteq X$
- e) $\{\} \subseteq X$ f) $\{\} \in X$ g) $\{1, 2, 3\} \subseteq X$ h) $\{1, 2, 3\} \subset X$
- i) $\{1, 2\} \subset X$ j) $|X| = 4$ k) $2 \subset X$

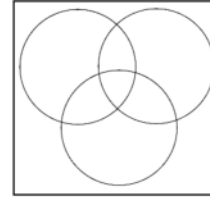
2. Use Venn diagrams to investigate truth or falsity of the following statement:

$$(A - B) \cap \overline{C} = (A \Delta B) - (B \cup C)$$



3. Consider universe $U = \{1, 2, 3, 4, 5\}$ and give one example of nonempty sets A, B and C that satisfies all of the following:

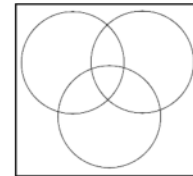
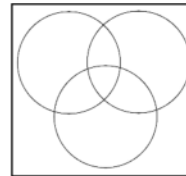
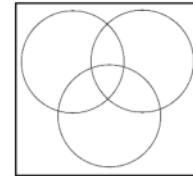
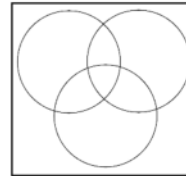
- $A \cap B = B \cap C$
- $A \subseteq B$
- $|A \cup B| = 3$
- $3 \in \overline{(A \cup B \cup C)}$
- $\{4, 5\} \subset C$



A = { }

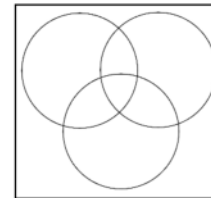
B = { }

C = { }



4. Consider universe $U = \{1, 2, 3, 4, 5\}$ and give one example of nonempty sets A, B and C that satisfies all of the following:

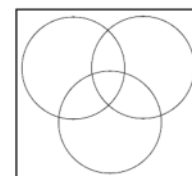
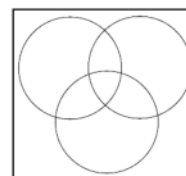
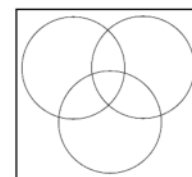
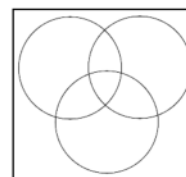
- $(A - B) \subseteq C$
- $|C - (A \cup B)| = 1$
- $\{3, 4, 5\} \subset \overline{A}$
- $A \cap C = A \cap B$



A = { }

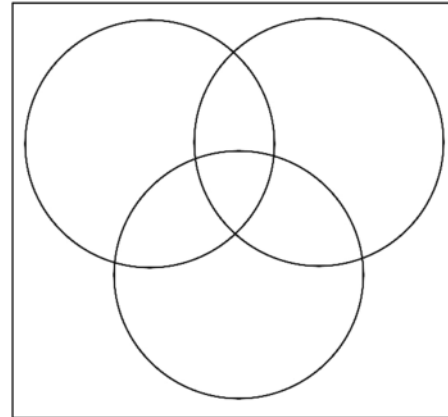
B = { }

C = { }



5. Provide one example of nonempty sets A, B and C that satisfies all of the following:

- $A \subseteq (B \Delta C)$
- $(\overline{A} \cap B \cap \overline{C}) = \{1\}$
- $((C - B) - \overline{(A \cup B)}) = \{2, 3\}$
- $|(A \cap C) \Delta (B \cup C)| = 3$
- $4 \notin ((A \Delta B) \Delta \overline{C})$



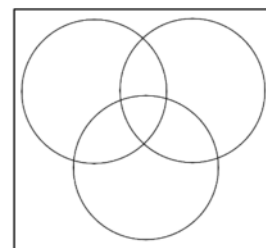
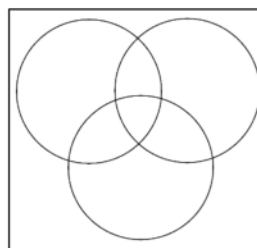
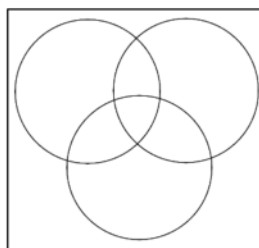
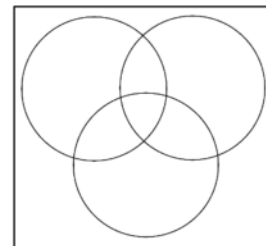
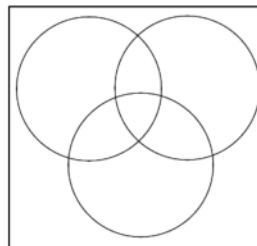
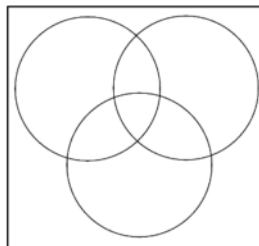
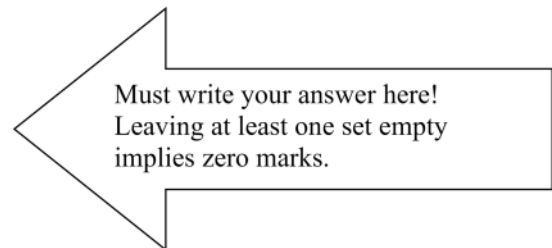
Universe $U = \{1, 2, 3, 4, 5, 6\}$.

Write your answer:

$A = \{ \quad \quad \quad \}$

$B = \{ \quad \quad \quad \}$

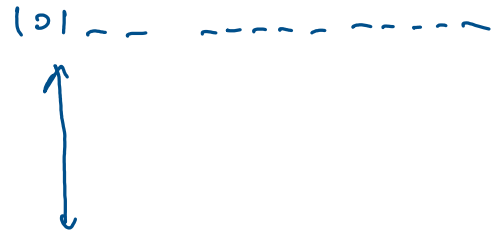
$C = \{ \quad \quad \quad \}$



6. What is the probability that a randomly selected binary string of length 15:

a) start with 101?

$$P_r = \frac{2^{12}}{2^{15}}$$



b) start with 101 or start with 0011?

$$P_r = \frac{2^{12} + 2^{11}}{2^{15}}$$

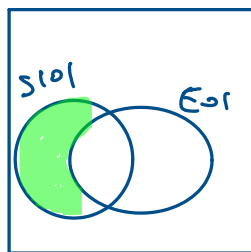


c) start with 101 or end with 01?

$$P_r = \frac{2^{12} + 2^{13} - 2^{10}}{2^{15}}$$

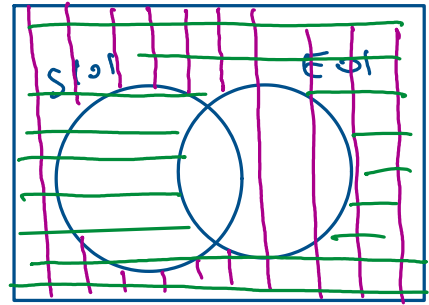


d) start with 101 but do not end with 01?



$$P_r = \frac{2^{12} - 2^{10}}{2^{15}}$$

e) do not start with 101 or do not end with 01?



$$P_r = \frac{2^{15} - 2^{10}}{2^{15}}$$

$$\text{NOT } S101 \text{ OR NOT } E01 = \text{NOT} \left(\underbrace{S101 \text{ AND } E01}_{\frac{2^{10}}{2^{15}}} \right)$$

f) do not start with 101 and do not end with 01?

NOT S101 AND NOT E01

$$1 - \frac{2^{10}}{2^{15}}$$

$$\text{NOT} \left(\underbrace{S101 \text{ OR } E01}_{\frac{2^{12} + 2^{13} - 2^{10}}{2^{15}}} \right)$$

$$= 1 - \frac{2^{12} + 2^{13} - 2^{10}}{2^{15}}$$

7. A student visits an arcade each day after school (Monday to Friday) and plays one game of either Pac-Man, Tetris, or Donkey Kong, but makes sure that each of the three types are played in the first three days of the week.

What is the probability that he plays Pac-Man on consecutive days of a given school week?

$$Pr = \frac{10}{3! (3)^2}$$

M T W R F
3 2 1 3 3
3!

P T D

M T W R F

→ P (2) (1) P P
(2) P (1) P P
(1) (1) P P (3)

2
2
6 } = 10

Additional Questions

Example. A student visits an arcade each day after school and plays one game of either Pac-Man, Tetris, or Donkey Kong. What is the probability that he plays each of the three types at least once during a given school week?

$$p_r = \frac{3 \cdot \frac{5!}{3!} + 3 \cdot \frac{5!}{2!2!}}{3^5}$$

$$\begin{array}{l} PPP T D \leftarrow \frac{5!}{3!} \\ TTT P D \leftarrow \frac{5!}{3!} \\ DDD P T \leftarrow \frac{5!}{3!} \end{array} \left\{ 3 \cdot \frac{5!}{3!} \right.$$

$$PPTTD \quad \frac{5!}{2!2!}$$

$$\begin{array}{cccccc} H & T & W & R & F \\ x & x & x & & \\ x & x & & x & \end{array}$$

Theorem: ~~If A, B, C are finite sets, then~~

$$\del{|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|}.$$